The type of bi-invertible maps is equivalent to the type of half adjoint equivalences

The Univalent Foundations Program

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Editorial note. The counted statement is the author cor environment at source lines 405-407 and its proof environment opens on line 408, the next line; the pairing is by adjacency and the retained proof range 408-411 is the whole environment.

Editorial note. The book label-number dump gives this label as 4.3.3 (env cor, chapter 4, section 3, third counter value), so the counters chapter 4, section 3 and thm 2 (zero-based) are set so that the item prints with the book number.

Editorial note. One label the proof cites is declared in external_references as a non-load-bearing navigation pointer: lem:equiv-iff-hprop in this chapter, the criterion that a map between mere propositions with a section is an equivalence; it is counted by its own card of this lane.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: bi-invertible maps are equivalent to half adjoint equivalences (author label thm:equiv-biinv-isequiv, book Corollary 4.3.3).

Allowed standard prerequisites (editorial).

- The three formulations of being an equivalence. A map is a half adjoint equivalence when it has a quasi-inverse with a coherence, and bi-invertible when it has a separate left and right inverse.. They are the two types compared by the statement.
- Implications between the formulations. A bi-invertible map is a quasi-inverse map, a quasi-inverse map is a half adjoint equivalence, and each of these steps is also available in the reverse direction.. They provide the two maps of the equivalence, one for each direction of the chain in the proof.
- Both formulations are mere propositions. Being a half adjoint equivalence and being bi-invertible are both mere propositions.. It is what allows the two implications to be upgraded to an equivalence by the cited criterion.

Source reference. Original lemma 3.3.3: sources/hott-logic/pinned-original-source.tex, source lines 294–294. Book Lemma 3.3.3, the criterion for maps between mere propositions, used to close the equivalence. Navigation pointer only: it is counted by its own card of this lane.

Corollary 4.3.3. *For any* $f : A \to B$ *we have* $\mathsf{biinv}(f) \simeq \mathsf{ishae}(f)$.

Proof. We have $\mathsf{biinv}(f) \to \mathsf{qinv}(f) \to \mathsf{ishae}(f)$ and $\mathsf{ishae}(f) \to \mathsf{qinv}(f) \to \mathsf{biinv}(f)$. Since both $\mathsf{ishae}(f)$ and $\mathsf{biinv}(f)$ are mere propositions, the equivalence follows from original lemma 3.3.3. □